

**DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 17, 2026
7. CONSERVATIVE SYSTEMS**

ROB TOMPKINS

Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- teamXX-YY.EXT

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

Team Activity

Three ways to find a conserved quantity:

- (1) Consider a system of the form $m\ddot{x} = F(x)$ where $F(x) = -\frac{dV}{dx}$ with $V(x)$ called the **potential energy**. $E = \frac{1}{2}m\dot{x}^2 + V$ is a conserved quantity for this system.

Example: Let $\ddot{x} = x^3$.

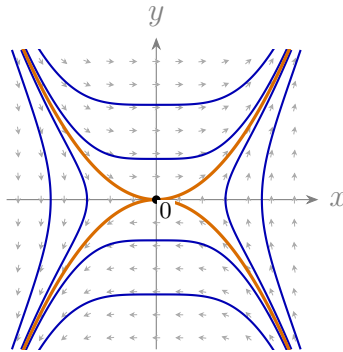
By introducing $y = \dot{x}$, this can be written as the system $\dot{x} = y, \dot{y} = x^3$. Then we have

$$V(x) = - \int x^3 dx = -x^4/4.$$

Try $E = \frac{1}{2}y^2 - x^4/4$. Is this conserved? (It should be).

$$\frac{dE}{dt} = y\dot{y} - x^3\dot{x} = yx^3 - x^3y = 0.$$

It is conserved!



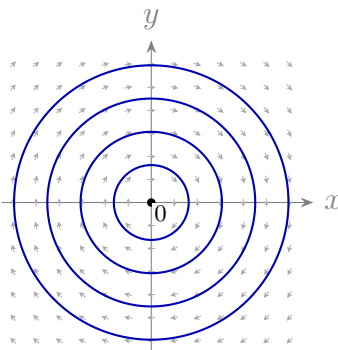
$\ddot{x} = x^3$: blue energy levels; orange $E = 0$ separatrices. Gray arrows show forward time.

Problem 1: Let $\ddot{x} = -kx$. Find E and check that it is conserved.

Solution. Set $y = \dot{x}$. Then $\dot{x} = y$ and $\dot{y} = -kx$, so (with unit mass)

$$V(x) = - \int (-kx) dx = \frac{kx^2}{2}, \quad \boxed{E(x, y) = \frac{1}{2}y^2 + \frac{k}{2}x^2}.$$

Along every solution, $\dot{E} = y\dot{y} + kx\dot{x} = y(-kx) + kxy = 0$. For $k > 0$, positive energy levels are ellipses $y^2 + kx^2 = 2E$, traversed clockwise, and the origin is a center. Their period is $2\pi/\sqrt{k}$. The conservation calculation holds for any constant k : $k = 0$ gives free motion, and $k < 0$ gives a saddle at the origin. $\square$



Harmonic oscillator, representative case $k = 1$.

- (2) A **phase curve** of a system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ is a curve $y = Y(x)$ (note that $\dot{y} = Y_x \dot{x}$) that obeys the differential equation $\frac{dY}{dx} = \frac{\dot{y}}{\dot{x}} = \frac{g(x, Y)}{f(x, Y)}$. Trajectories that start on such a curve stay on it. When $\frac{dy}{dx} = \frac{g(x, y)}{f(x, y)}$ is a separable differential equation, the process of solving it can also yield a conserved quantity.

Problem 2: A model of two species with exponential population growth and competition is given by

$$\begin{aligned}N_1' &= r_1 N_1 - \alpha_1 N_1 N_2 \\N_2' &= r_2 N_2 - \alpha_2 N_1 N_2.\end{aligned}$$

The model nondimensionalizes to

$$\begin{aligned}x' &= x - xy \\y' &= y(\rho - x).\end{aligned}$$

This nondimensionalization requires assuming that each population has its own units, so number of rabbits and number of sheep were each be assigned their own unknown constant in the nondimensionalization process.

- Using the nondimensional system, find an expression satisfied by most phase curves. Solve a separable differential equation to do this. You don't need to write it in the form $y = Y(x)$; an implicit relationship between x and y is fine.
- Argue that the quantity $H(x, y) = y - \ln y + \rho \ln x - x$ is almost always conserved. Which trajectories don't work with this function?

Solution. Assume positive growth and competition parameters, so $\rho > 0$. For example, $x = \alpha_2 N_1 / r_1$, $y = \alpha_1 N_2 / r_1$, $\tau = r_1 t$ give the stated system with $\rho = r_2 / r_1$. Wherever the division is valid,

$$\frac{dy}{dx} = \frac{y(\rho - x)}{x(1 - y)}, \quad \left(\frac{1}{y} - 1\right) dy = \left(\frac{\rho}{x} - 1\right) dx.$$

Integrating gives $\ln y - y = \rho \ln x - x + C$, or

$$\boxed{H(x, y) = y - \ln y + \rho \ln x - x = \text{constant.}}$$

Indeed, throughout the positive quadrant,

$$\dot{H} = \left(\frac{\rho}{x} - 1\right) x(1 - y) + \left(1 - \frac{1}{y}\right) y(\rho - x) = (\rho - x)(1 - y) + (y - 1)(\rho - x) = 0.$$

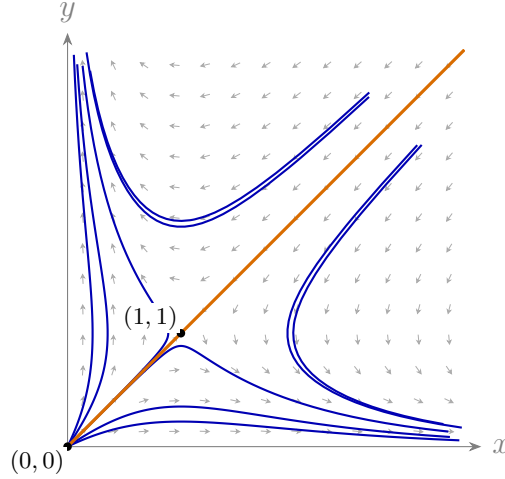
□

Problem 2, domain and phase portrait.

Solution. Thus H is conserved even where $x' = 0$ (including $y = 1$) and at the equilibrium $(\rho, 1)$; only the intermediate division fails there. A positive solution stays positive for as long as it exists, since each population satisfies a scalar equation of the form $u' = a(t)u$.

The displayed real logarithms exclude the invariant axes $x = 0$ and $y = 0$, including the origin. On these axes the nonzero population grows exponentially. Negative coordinates are also outside this H 's domain; replacing the logarithms by $\ln |x|$ and $\ln |y|$ extends the first integral to each open quadrant.

At $(\rho, 1)$ the Jacobian is $\begin{pmatrix} 0 & -\rho \\ -1 & 0 \end{pmatrix}$, with eigenvalues $\pm\sqrt{\rho}$, so coexistence is a saddle. For $\rho = 1$ its stable manifold is $y = x$ in the positive quadrant. $\square$



Competition for $\rho = 1$. The orange diagonal approaches the saddle; off it, one population eventually dominates.

- (3) A **Hamiltonian function**, $H(x, y)$ is an energy function. The associated dynamical system $\dot{x} = \frac{\partial H}{\partial y}$, $\dot{y} = -\frac{\partial H}{\partial x}$ is called a **Hamiltonian system**. In a Hamiltonian system, the Hamiltonian is a conserved quantity.

Solution. The chain rule proves conservation for every C^1 Hamiltonian:

$$\frac{dH}{dt} = H_x \dot{x} + H_y \dot{y} = H_x H_y - H_y H_x = 0.$$

$\square$

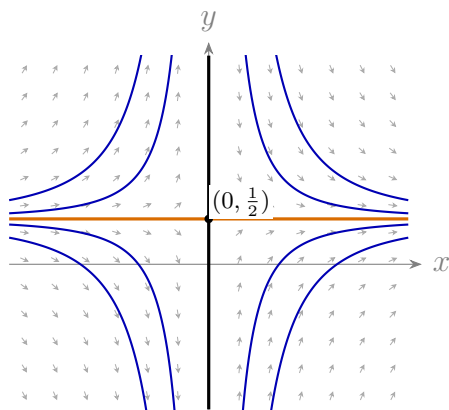
Problem 3: Let $\dot{x} = x^2, \dot{y} = -2xy + x$. Show this system is Hamiltonian.

Solution. Integrate $H_y = x^2$ with respect to y to obtain $H = x^2y + g(x)$. Then $-H_x = -2xy - g'(x)$ must equal $-2xy + x$, hence $g'(x) = -x$. Taking the additive constant to be zero gives

$$H(x, y) = x^2y - \frac{x^2}{2} = x^2 \left(y - \frac{1}{2} \right).$$

This has $H_y = x^2 = \dot{x}$ and $-H_x = -2xy + x = \dot{y}$. Directly, $\dot{H} = (2xy - x)x^2 + x^2(-2xy + x) = 0$.

For $x \neq 0$, the phase curves are $y = \frac{1}{2} + C/x^2$ and move toward increasing x . Every point on $x = 0$ is an equilibrium. The zero level consists of that entire line and the two moving half-lines $y = \frac{1}{2}, x \neq 0$; it is not a single trajectory. $\square$



Blue: nonzero levels of H . Orange: $y = \frac{1}{2}$. Black: the line of equilibria.

Problem 4: Consider the system

$$\begin{aligned}\dot{x} &= -\mu y + xy \\ \dot{y} &= \mu x + \frac{1}{2}(x^2 - y^2).\end{aligned}$$

Assume $\mu > 0$.

- (a) This system has a conserved quantity. Assume $\dot{x} = H_y$ and $\dot{y} = -H_x$ for an unknown function $H(x, y)$. If our assumption is correct then $\partial_x \dot{x} = -\partial_y \dot{y}$. Why is that?
- (b) Show that $\partial_x \dot{x} = -\partial_y \dot{y}$ for this system.
- (c) Construct a function $H(x, y)$ that is conserved for this system.
- (d) If time permits, begin the process of constructing a phase portrait for this system.

Solution. (a) If H is C^2 , equality of mixed partial derivatives gives $\partial_x \dot{x} = H_{yx} = H_{xy} = -\partial_y \dot{y}$.

- (b) Here $\partial_x [(-\mu + x)y] = y$ and $-\partial_y [\mu x + \frac{1}{2}(x^2 - y^2)] = y$.
- (c) Integrating $H_y = (x - \mu)y$ yields $H = \frac{1}{2}(x - \mu)y^2 + G(x)$. The second equation requires $-\frac{1}{2}y^2 - G'(x) = \mu x + \frac{1}{2}x^2 - \frac{1}{2}y^2$, so $G'(x) = -\mu x - \frac{1}{2}x^2$. Thus

$$H(x, y) = \frac{1}{2}(x - \mu)y^2 - \frac{\mu}{2}x^2 - \frac{1}{6}x^3.$$

Its Hamiltonian equations are exactly the given system, and therefore $\dot{H} = 0$.

- (d) Solving $(x - \mu)y = 0$ and $2\mu x + x^2 - y^2 = 0$ gives

$$(0, 0), \quad (-2\mu, 0), \quad (\mu, \sqrt{3}\mu), \quad (\mu, -\sqrt{3}\mu).$$

The Jacobian is $J(x, y) = \begin{pmatrix} y & x - \mu \\ \mu + x & -y \end{pmatrix}$. At the origin its eigenvalues are $\pm i\mu$. Since $D^2H(0, 0) = -\mu I$, H has a strict local maximum there: nearby energy levels are closed curves and the origin is a nonlinear center, traversed counterclockwise. At each of the other three equilibria the eigenvalues are $\pm\sqrt{3}\mu$, so all three are saddles.

□

Problem 4(d), completed phase portrait. Introduce $X = x/\mu$, $Y = y/\mu$, and $\tau = \mu t$. Then

$$\frac{dX}{d\tau} = (X - 1)Y, \quad \frac{dY}{d\tau} = X + \frac{1}{2}(X^2 - Y^2), \quad h = \frac{H}{\mu^3} = \frac{1}{2}(X - 1)Y^2 - \frac{1}{2}X^2 - \frac{1}{6}X^3.$$

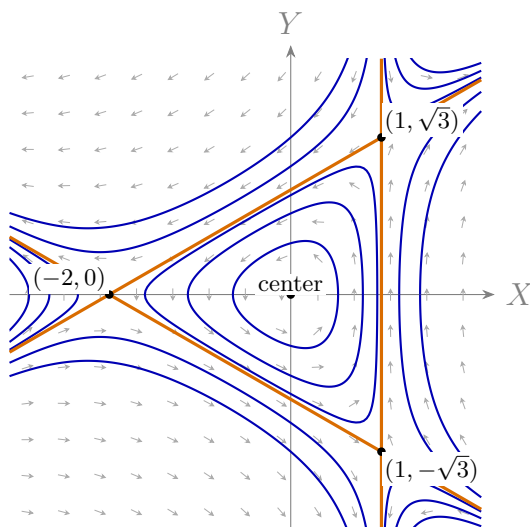
All three saddles lie on $h = -2/3$, and

$$h + \frac{2}{3} = -\frac{1}{6}(X - 1)((X + 2)^2 - 3Y^2).$$

Consequently the separatrix level is the union of the three straight lines

$$X = 1, \quad Y = \frac{X + 2}{\sqrt{3}}, \quad Y = -\frac{X + 2}{\sqrt{3}}.$$

Their finite connecting segments form a heteroclinic triangle around the center. Forward time goes from $(1, -\sqrt{3})$ to $(1, \sqrt{3})$, then to $(-2, 0)$, then back to $(1, -\sqrt{3})$; each saddle is approached only asymptotically. The exterior rays are also separatrix branches. Inside the triangle, the levels $-2/3 < h < 0$ are periodic orbits; outside it, level-set components are unbounded.



Normalized phase portrait for every $\mu > 0$. Orange: exact separatrices. Blue: sampled energy levels.

REVERSIBLE SYSTEMS:

A system $\dot{z} = F(z)$, with $z = (x, y)$, is **reversible** if the change of variables $t \mapsto -t$ and $(x, y) \mapsto R(x, y)$ with $R^2(x, y) = (x, y)$ yields the same system, i.e. the system is invariant to this change of variables.

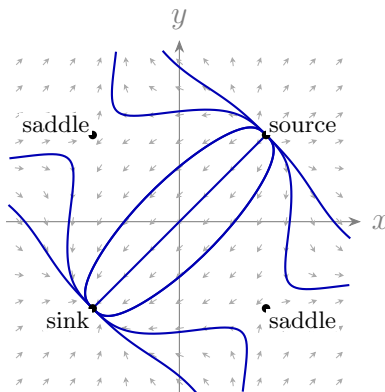
- examples of R : $R(x, y) = (x, -y)$ (mechanical systems, $m\ddot{x} = F(x)$) or $R(x, y) = (-x, -y)$. In 2D, every reflection about an axis through the origin satisfies $R^2 = I$; reversibility also requires $F(Rz) = -DR(z)F(z)$.
- For $R(x, y) = (x, -y)$, require f odd in y and g even in y .
- Time-reversal symmetry alone does not imply that a system has a conserved energy.
- Dissipative effects such as friction often break time-reversal symmetry in mechanical models.
- Similar to conservative systems, but not the same thing, e.g.

$$\dot{x} = y, \quad \dot{y} = x^3$$

is both conservative and reversible but

$$\dot{x} = -2 \cos x - \cos y, \quad \dot{y} = -2 \cos y - \cos x$$

is only reversible. (How can you tell it is not conservative from the phase portrait below?)



One 2π -periodic square; opposite edges are identified. Gray arrows give forward time.

Solution. For the mechanical example, $R(x, y) = (x, -y)$ satisfies $F(Rz) = -DRF(z)$; its conserved energy was found above. For the cosine system take $R(x, y) = (-x, -y)$. Because cosine is even, $F(-z) = F(z) = -DRF(z)$, proving reversibility.

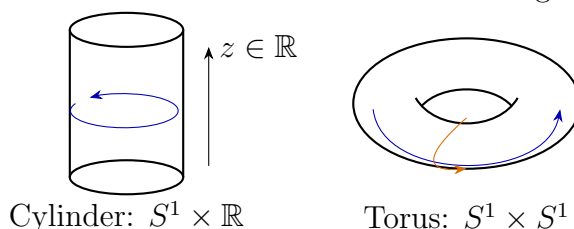
Its Jacobian is

$$J = \begin{pmatrix} 2 \sin x & \sin y \\ \sin x & 2 \sin y \end{pmatrix}.$$

At $(\pi/2, \pi/2)$ the eigenvalues are $3, 1$ (a source), and at $(-\pi/2, -\pi/2)$ they are $-3, -1$ (a sink). The mixed-sign equilibria have eigenvalues $\pm\sqrt{3}$ and are saddles. The sink explains the failure of an energy-level phase portrait: any continuous conserved function must have the same value throughout its basin, since $H(z) = \lim_{t \rightarrow \infty} H(\phi_t(z)) = H(\text{sink})$. Thus it cannot provide distinct regular energy curves near the sink. This is the relevant sense in which this reversible example is not conservative; it does not rule out local first integrals away from equilibria. $\square$

A **cylindrical phase space** arises when one coordinate can take on any value in $\mathbb{R}$ (the real numbers) while the other coordinate is an angle.

A **toroidal phase space** arises when two coordinates are angles.



Schematic surfaces. On the torus, both angular coordinates are taken modulo 2π .

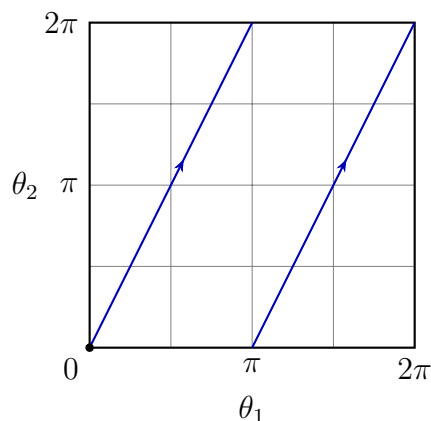
Problem 5: To get used to a phase space that is a torus, think about two oscillators that are not interacting (like the hour hand and the minute hand on a clock: they each go at their own pace, and that pace is constant).

- Let $\dot{\theta}_1 = 1$ and $\dot{\theta}_2 = 2$. If the oscillators each start at a phase angle of zero, so at the point $(0, 0)$, draw their trajectory onto the phase space. Use a square to represent the space. Will the pair of oscillators pass through $(0, 0)$ at some point?
- Now let $\dot{\theta}_1 = \pi$ and $\dot{\theta}_2 = 2\pi$. With an initial condition of $(0, 0)$, draw their trajectory onto the phase space. How is the trajectory different from the one in part a?
- Let $\dot{\theta}_1 = \pi$ and $\dot{\theta}_2 = \sqrt{2}\pi$. Assume the oscillator pair again starts at $(0, 0)$. The first oscillator will return to a phase of zero at time 2, time 4, etc. When does the second oscillator return to a phase of zero? Will the pair pass through $(0, 0)$ at some point?

Solution. The phase space is $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$. In its square representation, identify left with right and bottom with top; in particular, all four corners represent $(0, 0)$.

- The solution is $(\theta_1, \theta_2) = (t, 2t) \pmod{2\pi}$. The lifted line has slope 2. In the square it runs from $(0, 0)$ to $(\pi, 2\pi)$, reenters at $(\pi, 0)$, and reaches $(2\pi, 2\pi) \sim (0, 0)$. Simultaneous return requires $t = 2\pi m$ and $2t = 2\pi n$; the least positive solution is $T = 2\pi$.
- Now $(\theta_1, \theta_2) = (\pi t, 2\pi t) \pmod{2\pi}$. The geometric path and its direction are identical to part (a), but the speed is multiplied by π and the least positive period is $T = 2$.

□



Parts (a) and (b): the same closed orbit, with different traversal times.

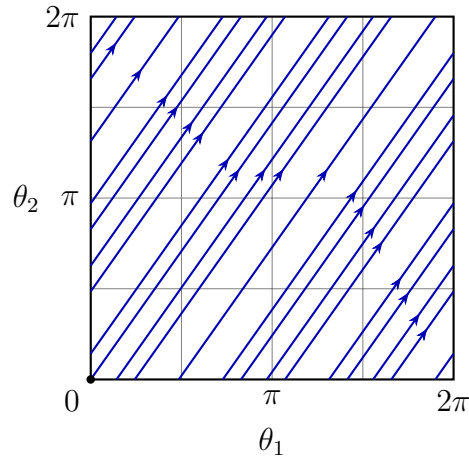
Problem 5(c): irrational frequency ratio.

Solution. Here $(\theta_1, \theta_2) = (\pi t, \sqrt{2}\pi t) \pmod{2\pi}$. The first oscillator returns to zero at $t = 2m$, $m \in \mathbb{Z}_{>0}$. The second returns when $\sqrt{2}\pi t = 2\pi n$, so

$$t = \sqrt{2}n, \quad n \in \mathbb{Z}_{>0}.$$

A common positive return would require $2m = \sqrt{2}n$, hence $\sqrt{2} = n/m$, contradicting irrationality. Therefore the pair *never returns exactly* to $(0, 0)$ after $t = 0$.

The lifted trajectory is a line of slope $\sqrt{2}$. On the torus it is nonperiodic and dense. To see density, at times $t = 2m + \alpha/\pi$ the first angle is any prescribed α , while the second angles differ by $2\pi\sqrt{2}$ modulo 2π . Irrational rotation makes these angles dense in the circle, so the orbit meets every open rectangle in the torus. It comes arbitrarily close to its initial state without returning exactly. $\square$



Part (c): a finite segment, $0 \leq t \leq 18$, of the dense orbit. Opposite-edge exits and entries join continuously on the torus.